

Contents

Sample 1- Ultra-short baseline (USBL) acoustic tracking system for ROVs used by government agencies (research, maritime) 2

Sample 2: Ultrasonic Technology Instruments to deliver highly accurate measurements with no moving parts to wear or recalibrate in maritime, meteorology, and urban environment 3

Sample 3 [REDACTED] Chemical Concentration Meter 5

Introduction

Please refer to quick research and a prompt comparison named :Sample 3 ██████████ Chemical Concentration Meter: with few of its competitors along with two other Instruments comparison in advance application listed as sample 1 and sample 2.

It is prepared to have your attention and discuss how I can bring great value to the table due to my relevant experience. As that is what I do for my current employer to propose best value solutions in the Analyzer, Instruments, ██████████, sensors, and measurement/testing equipment domain.

Sample 1- Ultra-short baseline (USBL) acoustic tracking system for ROVs used by government agencies (research, maritime)				
USBL (ultra-short baseline, also known as SSBL for super short base line) is a method of underwater acoustic positioning. A USBL system consists of a transceiver, which is mounted on a pole under a ship, and a transponder or responder on the seafloor, on a towfish, or on an ROV. A computer, or "topside unit", is used to calculate a position from the ranges and bearings measured by the transceiver.				
Specs comparison with only requirements (models are deleted on purpose)				
	██████████	██████████	██████████	██████████ ██████████
Requirement				
1. Slant range of 3000 m	Meets	Meets (up to 5000 m)	Meets (up to 3000 m)	Meets (up to 2000 m)
2. Acoustic coverage of 90 degrees	Meets	Meets	Meets	Meets
3. Positional error of 0.2%	Meets	Meets	Meets	Meets
4. Broadband acoustic reply	Meets	Meets	Meets	Meets
5. ██████████ output	Meets	Meets	Meets	Meets
6. Depth telemetry support	Meets	Meets	Meets	Meets
7. Pressure sensor compatibility	Meets	Meets	Meets	Meets
8. Multi-beacon tracking	Meets	Meets	Meets	Meets
9. Low voltage DC responder trigger	Meets	Meets	Meets	Meets
10. Surface data processor components in a 19-inch rack unit	Meets	Meets	Meets	Meets
11. Offline software licensing	Meets	Meets	Meets	Meets
12. Pitch/roll/heading aiding	Meets	Meets	Meets	Meets
13. Cabling connecting transceiver to surface data processor is at least 30 m long	Meets	Meets	Meets	Meets
14. End user can upload a sound velocity profile	Meets	Meets	Meets	Meets
15. USBL transceiver can be pole mounted	Meets	Meets	Meets	Meets
16. Contractor provides technical support	Meets	Meets	Meets	Meets
17. Contractor provides printed and PDF copies of operations manuals	Meets	Meets	Meets	Meets
18. Beacon rated to a depth of a minimum of 1000 m	Meets	Meets	Meets	Meets
19. Beacon diameter must not exceed 90 mm and a length must not exceed 300 mm	Partially meets	Partially meets	Partially meets	Partially meets
Additional Specifications:				
Feature	██████████	██████████	██████████	██████████ ██████████
Frequency	Varies (e.g., 40 kHz, 75 kHz)	Varies	Varies	Varies
Data Rate	Up to 256 kbaud	Up to 1 Mbaud	Up to 1 Mbaud	Up to 512 kbaud
Power Consumption	Varies	Varies	Varies	Varies
Dimensions	Varies	Varies	Varies	Varies

Sample 2: Ultrasonic Technology Instruments to deliver highly accurate measurements with no moving parts to wear or recalibrate in maritime, meteorology, and urban environment

r #	Parameter	Requirement	MODEL OMITTED	[REDACTED] company		[REDACTED] company		http://[REDACTED].com		[REDACTED]/distributors/		Meets/Exceeds	Notes
				[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	MODEL OMITTED	MODEL OMITTED	MODEL OMITTED	MODEL OMITTED		
1	Wind Speed	Range 0-60 m/s, accuracy ±3% at 10 m/s	Range: 0-60 m/s, Accuracy: ±3% at 10 m/s	Range: 0-60 m/s, Accuracy: ±2% at 10 m/s	Range: 0-50 m/s, Accuracy: ±3% at 10 m/s	Range: 0-60 m/s, Accuracy: ±3% at 10 m/s	Range: 0-50 m/s, Accuracy: ±2% at 10 m/s	Range: 0-60 m/s, Accuracy: ±4% at 10 m/s	Range: 0-50 m/s, Accuracy: ±2% at 10 m/s	Range: 0-50 m/s, Accuracy: ±2% at 10 m/s	Range: 0-50 m/s, Accuracy: ±2% at 10 m/s	Meets	All meet the range; [REDACTED] and [REDACTED] exceed accuracy requirement
2	Wind Direction	Range 0-360°, accuracy ±3.0° at 10 m/s	Range: 0-360°, Accuracy: ±3.0° at 10 m/s	Range: 0-360°, Accuracy: ±2.5° at 10 m/s	Range: 0-360°, Accuracy: ±3.0° at 10 m/s	Range: 0-360°, Accuracy: ±2.5° at 10 m/s	Range: 0-360°, Accuracy: ±4.0° at 10 m/s	Range: 0-360°, Accuracy: ±2.5° at 10 m/s	Range: 0-360°, Accuracy: ±3.0° at 10 m/s	Range: 0-360°, Accuracy: ±3.0° at 10 m/s	Range: 0-360°, Accuracy: ±3.0° at 10 m/s	Meets	All meet the range; [REDACTED] and [REDACTED] exceed accuracy requirement
3	Air Pressure	Range 500-1100 hPa, accuracy ±0.5 hPa at 0-30 °C	Range: 500-1100 hPa, Accuracy: ±0.5 hPa	Range: 500-1100 hPa, Accuracy: ±0.3 hPa	Range: 500-1050 hPa, Accuracy: ±0.5 hPa	Range: 500-1100 hPa, Accuracy: ±0.3 hPa	Range: 500-1200 hPa, Accuracy: ±1.0 hPa	Range: 500-1100 hPa, Accuracy: ±0.5 hPa	Range: 500-1100 hPa, Accuracy: ±0.3 hPa	Range: 500-1100 hPa, Accuracy: ±0.3 hPa	Range: 500-1100 hPa, Accuracy: ±0.3 hPa	Meets/Exceeds	[REDACTED] exceeds range requirement
4	Temperature	Range -52 to +60 °C, accuracy ±0.3 °C at 20 °C	Range: -52 to +60 °C, Accuracy: ±0.3 °C	Range: -50 to +60 °C, Accuracy: ±0.2 °C	Range: -52 to +60 °C, Accuracy: ±0.5 °C	Range: -50 to +60 °C, Accuracy: ±0.3 °C	Range: -50 to +65 °C, Accuracy: ±0.5 °C	Range: -52 to +60 °C, Accuracy: ±0.3 °C	Range: -50 to +60 °C, Accuracy: ±0.2 °C	Range: -50 to +60 °C, Accuracy: ±0.2 °C	Range: -50 to +60 °C, Accuracy: ±0.2 °C	Meets	All meet the range; [REDACTED] and [REDACTED] exceed accuracy requirement
5	Precipitation	Acoustic measurement, range 0-200 mm/h	Acoustic measurement, Range 0-200 mm/h	Optical measurement, Range 0-150 mm/h	Acoustic measurement, Range 0-200 mm/h	Optical measurement, Range 0-200 mm/h	Acoustic measurement, Range 0-200 mm/h	Acoustic measurement, Range 0-200 mm/h	Optical measurement, Range 0-200 mm/h	Optical measurement, Range 0-200 mm/h	Optical measurement, Range 0-200 mm/h	Meets/Does Not Meet	[REDACTED] meet; [REDACTED] and [REDACTED] do not
6	Humidity	Range 0-100% RH, accuracy ±3% RH at 0-90% RH	Range: 0-100% RH, Accuracy: ±3% RH at 0-90% RH	Range: 0-100% RH, Accuracy: ±2% RH at 0-85% RH	Range: 0-100% RH, Accuracy: ±4% RH at 0-95% RH	Range: 0-100% RH, Accuracy: ±3% RH at 0-90% RH	Range: 0-100% RH, Accuracy: ±4% RH at 0-95% RH	Range: 0-100% RH, Accuracy: ±3% RH at 0-90% RH	Range: 0-100% RH, Accuracy: ±2% RH at 0-85% RH	Range: 0-100% RH, Accuracy: ±2% RH at 0-85% RH	Range: 0-100% RH, Accuracy: ±2% RH at 0-85% RH	Meets/Does Not Meet	[REDACTED] do not meet accuracy requirement

7	Integrated Sensors	Air Pressure, Temperature, and Humidity sensors integrated into one module; calibrated and certified with SI traceability	Integrated, Calibrated and Certified with SI Traceability	Integrated, Calibrated and Certified with SI Traceability	Integrated, Calibrated and Certified with SI Traceability	Integrated, Calibrated and Certified with SI Traceability	Integrated, Calibrated and Certified with SI Traceability	Integrated, Calibrated and Certified with SI Traceability	Integrated, Calibrated and Certified with SI Traceability	Integrated, Calibrated and Certified with SI Traceability	Meets	All meet requirements
8	Communication Digital Outputs	SDI-12, RS-232, RS-485, [REDACTED]	SDI-12, RS-232, RS-485, [REDACTED]	SDI-12, RS-232, RS-485, [REDACTED]	SDI-12, RS-232, RS-485, [REDACTED]	SDI-12, RS-232, RS-485, [REDACTED]	SDI-12, RS-232, RS-485, [REDACTED]	SDI-12, RS-232, RS-485, [REDACTED]	SDI-12, RS-232, RS-485, [REDACTED]	SDI-12, RS-232, RS-485, [REDACTED]	Meets	All meet requirements
9	Communication Protocols	SDI-12 v1.3, Modbus RTU, ASCII, NMEA 0183 v3.0	SDI-12 v1.3, Modbus RTU, ASCII, NMEA 0183 v3.0	SDI-12 v1.3, Modbus RTU, ASCII, NMEA 0183 v3.0	SDI-12 v1.3, Modbus RTU, ASCII, NMEA 0183 v3.0	SDI-12 v1.3, Modbus RTU, ASCII, NMEA 0183 v3.0	SDI-12 v1.3, Modbus RTU, ASCII, NMEA 0183 v3.0	SDI-12 v1.3, Modbus RTU, ASCII, NMEA 0183 v3.0	SDI-12 v1.3, Modbus RTU, ASCII, NMEA 0183 v3.0	SDI-12 v1.3, Modbus RTU, ASCII, NMEA 0183 v3.0	Meets	All meet requirements
10	Operating Voltage	6-24 V DC	6-24 V DC	6-24 V DC	6-24 V DC	6-24 V DC	6-24 V DC	6-24 V DC	6-24 V DC	6-24 V DC	Meets	All meet requirements
11	Heating Voltage	12-24 V DC, 12-17 V AC	12-24 V DC	12-24 V DC	12-24 V DC	12-24 V DC	12-24 V DC	12-24 V DC	12-24 V DC	12-24 V DC	Meets	All meet requirements
12	Operating Temperature	-52 to +60 °C	-52 to +60 °C	-52 to +60 °C	-52 to +60 °C	-52 to +60 °C	-52 to +60 °C	-52 to +60 °C	-52 to +60 °C	-52 to +60 °C	Meets	All meet requirements
13	Humidity	0-100% RH	0-100% RH	0-100% RH	0-100% RH	0-100% RH	0-100% RH	0-100% RH	0-100% RH	0-100% RH	Meets	All meet requirements
14	IP Rating	IP65/IP66 with mounting kit	IP66 with Mounting Kit	IP65/IP66 with Mounting Kit	IP66 with Mounting Kit	IP65/IP66 with Mounting Kit	IP66 with Mounting Kit	IP66 with Mounting Kit	IP66 with Mounting Kit	IP66 with Mounting Kit	Meets	All meet requirements
15	Connector	8 pin M12 connector with bushing and grounding accessories	8 pin M12 Connector	8 pin M12 Connector	8 pin M12 Connector	8 pin M12 Connector	8 pin M12 Connector	8 pin M12 Connector	8 pin M12 Connector	8 pin M12 Connector	Meets	All meet requirements
16	Bird Spike Kit	Included	Included	Included	Included	Included	Included	Included	Included	Included	Meets	All meet requirements
17	Mounting Kit	Included	Included	Included	Included	Included	Included	Included	Included	Included	Meets	All meet requirements
18	Maximum Dimensions	Diameter: 4.5", Height: 7"	Diameter: 4.5", Height: 7"	Diameter: 4.5", Height: 7"	Diameter: 4.5", Height: 7"	Diameter: 4.5", Height: 7"	Diameter: 4.5", Height: 7"	Diameter: 4.5", Height: 7"	Diameter: 4.5", Height: 7"	Diameter: 4.5", Height: 7"	Meets	All meet requirements
19	IEC Compliance	IEC 60945, DNV GL Type Examination Certificate	IEC 60945, DNV GL Certificate	IEC 60945, DNV GL Certificate	IEC 60945, DNV GL Certificate	IEC 60945, DNV GL Certificate	IEC 60945, DNV GL Certificate	IEC 60945, DNV GL Certificate	IEC 60945, DNV GL Certificate	IEC 60945, DNV GL Certificate	Meets	All meet requirements
20	EN Compliance	EN 61326-1, industrial environment CISPR 32 /	EN 61326-1, CISPR 32 / EN 55032, Class B	EN 61326-1, CISPR 32 / EN 55032, Class B	EN 61326-1, CISPR 32 / EN 55032, Class B	EN 61326-1, CISPR 32 / EN 55032, Class B	EN 61326-1, CISPR 32 / EN 55032, Class B	EN 61326-1, CISPR 32 / EN 55032, Class B	EN 61326-1, CISPR 32 / EN 55032, Class B	EN 61326-1, CISPR 32 / EN 55032, Class B	Meets	All meet requirements

		EN 55032, Class B										
2 1	Compliance	IEC 60068-2-1, 2, 6, 14, 30, 31, 78 IEC 60529, VDA 621-415	IEC 60068-2-1, 2, 6, 14, 30, 31, 78, 60529, VDA 621-415	IEC 60068-2-1, 2, 6, 14, 30, 31, 78, 60529, VDA 621-415	IEC 60068-2-1, 2, 6, 14, 30, 31, 78, 60529, VDA 621-415	IEC 60068-2-1, 2, 6, 14, 30, 31, 78, 60529, VDA 621-415	IEC 60068-2-1, 2, 6, 14, 30, 31, 78, 60529, VDA 621-415	IEC 60068-2-1, 2, 6, 14, 30, 31, 78, 60529, VDA 621-415	IEC 60068-2-1, 2, 6, 14, 30, 31, 78, 60529, VDA 621-415	IEC 60068-2-1, 2, 6, 14, 30, 31, 78, 60529, VDA 621-415	Meets	All meet requirements

Academically I am engineer with strong experience in Autoamtion Solutions (SCADA, RTU, ESD,), LV/MV,my capability. Notes on Comparisons:

Sample 3 ██████████ Chemical Concentration Meter

Measurement Principle: All listed products use different measurement technologies suited for their target applications. ██████████ employs ultrasonic technology, while ██████████ and ██████████ use conductivity and optical methods.

Measuring Range: ██████████ has a specified range of 0-100 g/L, while ██████████'s range is application-specific, which can vary significantly based on the chemicals being measured. This may require further investigation.

Operating Conditions: ██████████'s products are designed for extremely harsh environments, making them suitable for a wider variety of chemical processes. ██████████ provides a good temperature range, but users should verify suitability for their specific applications.

Material Construction: ██████████ emphasizes corrosion resistance, a vital feature for chemical applications. Other competitors also provide robust materials but may have specific configurations that could be advantageous based on the chemical nature.

Power Supply and Output Signal: The electrical specifications are largely comparable, but specific models from competitors may offer more variety, especially in terms of communication protocols.

User Interface: ██████████ provides a straightforward display for real-time data, whereas ██████████ and others may offer more advanced interfaces, including touchscreens.

Applications: While ██████████ targets chemical, food, and wastewater applications, other manufacturers may have specialized offerings in the semiconductor sector, reflecting market demands.

Model Availability: ██████████ lists specific models, while some competitors offer broader series with varying features tailored to specific applications. This could affect user choice based on their requirements.

Feature	██████████ Chemical Concentration Meter	██████████ Concentration Measurement	██████████ Series	██████████	██████████ Chemical Concentration Monitor
Measurement Principle	Ultrasonic technology	4-electrode conductivity measurement	Optical measurement	Optical measurement	Inline and offline chemical concentration measurement
Measuring Range	0 – 100 g/L	Specific to application; designed for various chemicals	Measures various chemical concentrations (dependent on model)	Up to 4 components, specific ranges available	Varies by application
Operating Temperature	-20 °C to +60 °C	Designed for harsh conditions; specifics vary by model	Typically -20 °C to +50 °C	-10 °C to +60 °C	Varies; check specific models
Pressure Resistance	Up to 6 bar	Pressure tested for extreme conditions; varies by application	Specific to model	Up to 5 bar	Varies by model
Installation	Inline installation	Inline; minimal calibration required	Inline or offline options	Inline installation	Inline, easy to integrate
Material of Construction	Corrosion-resistant materials	High-quality, corrosion-resistant materials	Robust materials for chemical resistance	Sealed structure, fluoroplastic coating	High-purity materials for chemical resistance
Power Supply	24 VDC or 100-240 VAC	Standard voltage options	24 VDC or AC options	100-240 VAC	Varies by model
Output Signal	4-20 mA or ██████████	4-20 mA or digital outputs available	4-20 mA or digital communication	4-20 mA	Analog or digital outputs
Calibration	Factory calibrated	Factory calibrated	Factory calibrated	Factory calibrated	Factory calibrated
User Interface	Display with real-time data	User-friendly with monitoring features	User-friendly interface	Touchscreen or user-friendly interface	User-friendly, integrated into process controls
Maintenance	Low maintenance requirements	Low maintenance, designed for long life	Low maintenance	Low maintenance	Low maintenance
Applications	Chemical industry, wastewater treatment, food & beverage	Harsh chemical environments, various industrial applications	Semiconductor manufacturing and critical processes	Semiconductor, LCD production, general industrial uses	Semiconductor and high-tech industries
Models Available	██████████, ██████████, ██████████, ██████████	██████████ series	██████████, ██████████, ██████████, ██████████	██████████, ██████████	Various models based on specific chemical requirements



General Specs (applicable to all models):

Feature	Description
Measurement Method	Speed of Sound
Temperature Range	5°C to 50°C (41°F to 122°F)
Maximum Pressure	4 bar
Power Supply	24 VDC (18-32V), 8 Watt
Output	Modbus RTU over [REDACTED]
Data Logging	Continuous, retrievable via USB stick (max. 65,535 entries)
Cable Glands	2x M16X1.5 for Ø 2-6 mm cable
Ambient Temperature	5°C to 60°C (41°F to 140°F)
Humidity	< 95% at 40°C (non-condensing)
Display	240x128 dots (WxH), 5 colors
Process Connection	[REDACTED]
Protection Rating	IP65, NEMA 4
Dimensions (W x H x D)	200 x 225 x 142 mm
Weight	Approx. 11 kg

[REDACTED] Chemical Concentration Meter ([REDACTED]) Specs

Model	Number of Chemicals Measured	Concentration Range	Concentration Accuracy	Temperature Accuracy	Other Features
[REDACTED]	1	-	Up to ±0.001 WT%	Up to 50°C	-
[REDACTED]	2	-	Up to ±0.001 WT%	Up to 50°C	-
[REDACTED] series	1, 2, or 3	-	Up to ±0.001 WT%	Up to 50°C	-
[REDACTED]	1 (COD in sugar-based wastewater)	-	Up to ±0.001 WT%	Up to 50°C	-